TABLE OF SPECIFICATIONS & QUESTION BANK

ITE241 — Advanced Digital Electronics
First Quarter · S.Y. 2026–2027 · MSU-IIT IDS Online Workspace

This assessment package uses the same material source and coverage as IT10: Arduino Part 2, Resistor Color Code, and Programming with Arduino. ITE241's exam is multiple-choice only.

> **Assessment format:** Three quizzes — 20 items each, 10 minutes each; one essay assessment — 2 items, 10 minutes; exam — 50 multiple-choice items, 60 minutes. Essays are graded using case-insensitive rubric keywords and are excluded from the exam.

Table of Specifications

Assessment / Content Area	Items	Coverage and Cognitive Focus	Format / Time
Quiz — Arduino Part 2	20	Arduino functions, libraries, sensors, actuators, potentiometers, and related applications. Remembering, understanding, applying.	Multiple choice · 10 min
Quiz — Resistor Color Code	20	Four-band and five-band codes, digit and multiplier values, tolerance, and decoding resistor values. Remembering, understanding, applying.	Multiple choice · 10 min
Quiz — Programming with Arduino	20	Guide structure, Arduino environment, basic electronics, programming examples, and code concepts. Remembering, understanding, applying.	Multiple choice · 10 min
Essay Assessment — Electronics Applications	2	Explain a sensor–program–actuator application; decode resistor bands and explain multimeter verification. Understanding, applying.	Short essay · 10 min
First Quarter Examination	50	Integrated coverage of all three source areas. Recall, conceptual understanding, code interpretation, resistor decoding, and circuit application.	Multiple choice only · 60 min
Total unique assessment slots	**112**	**Three 20-item quizzes, two essay prompts, and 50 multiple-choice exam items.**	

Exam blueprint

Content domain	Items	Primary cognitive demand
Arduino Part 2: functions, libraries, sensors and actuators	17	Remembering, understanding, applying
Resistor Color Code: bands, tolerance, value decoding	17	Understanding, applying, analyzing
Programming with Arduino: environment, syntax and examples	16	Remembering, understanding, applying

Content domain	Items	Primary cognitive demand
Total	**50**	**Multiple choice**

The domain allocation is a balanced ITE241 blueprint over the same IT10 source coverage. The exam bank is populated from available multiple-choice questions in that source bank; the 50-item draw is limited to those MC questions.

Question Bank

Arduino Part 2

Review Arduino setup and loop functions, libraries, sensors and actuators, analog inputs, potentiometers, single-board devices, and practical circuit tasks.

Resistor Color Code

Review band order, significant digits, multipliers, tolerance bands, 4-band and 5-band decoding, and verification with a multimeter.

Programming with Arduino

Review the guide's five-part organization, development environment, introductory electronics, programming structures, and examples.

Essay prompts and grading

1. Explain how an Arduino sensor and an actuator work together in a practical circuit. Give one example and describe the role of the program.
2. Describe how you would identify a resistor value from its color bands and verify that value with a multimeter.

Essay keyword matching is case-insensitive. Each response is evaluated against the rubric concepts for its prompt.